11.1.2] Sum of Digits using Recursion

Algorithm :-

Step 1: Start the program

Step 2: Define a function sum_of_digits_recursive(n)

Step 3: Check the base condition

→ If n == 0, return 0

Step 4: Otherwise (recursive case)

Step 4.1: Find last digit

→ digit = n % 10

Step 4.2: Call the function again with remaining number

→ sum_of_digits_recursive(n // 10)

Step 4.3: Add digit to recursive result

→ return digit + sum_of_digits_recursive(n // 10)

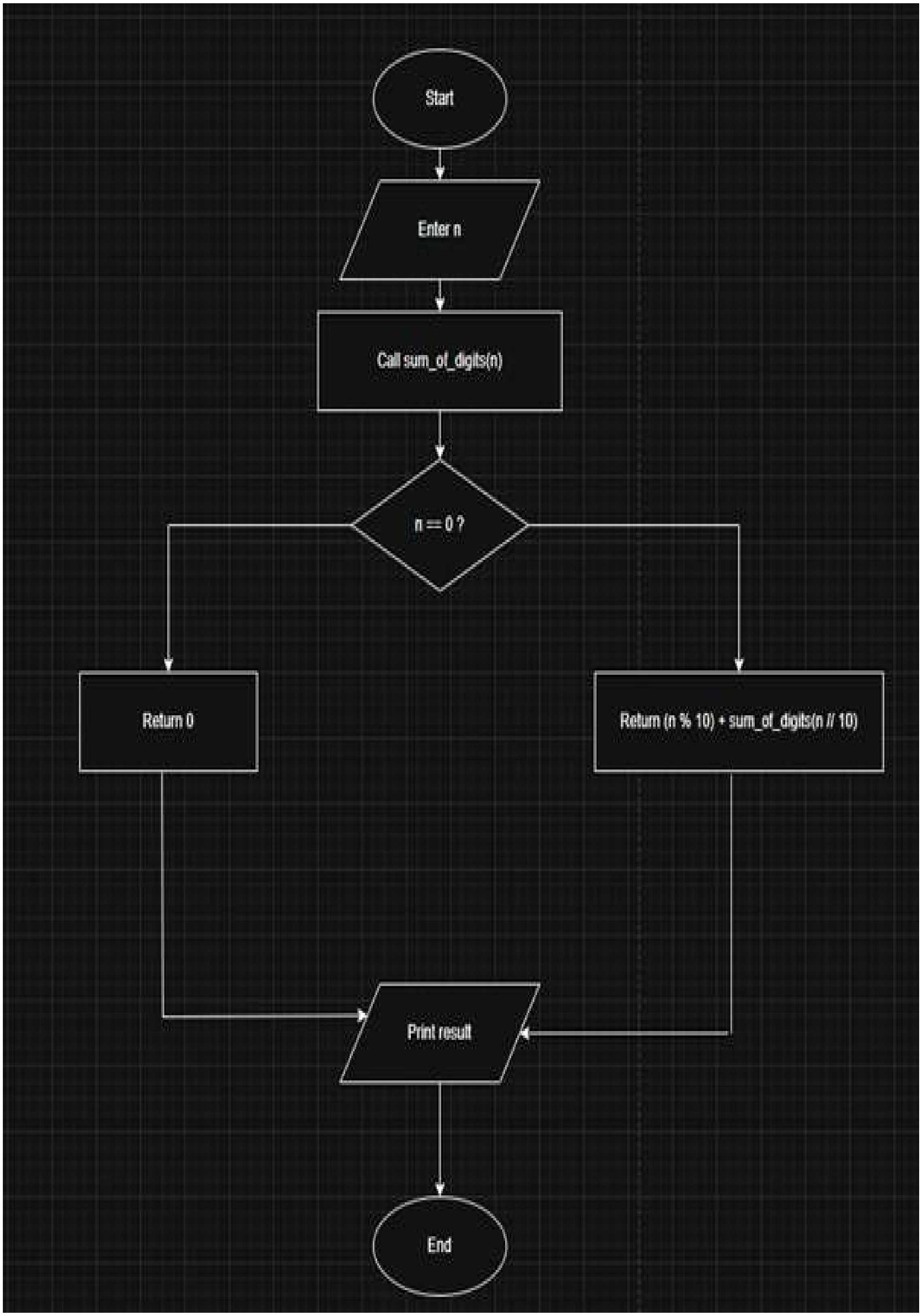
Step 5: Take input number from user

Step 6: Call the recursive function and store result

Step 7: Print the result

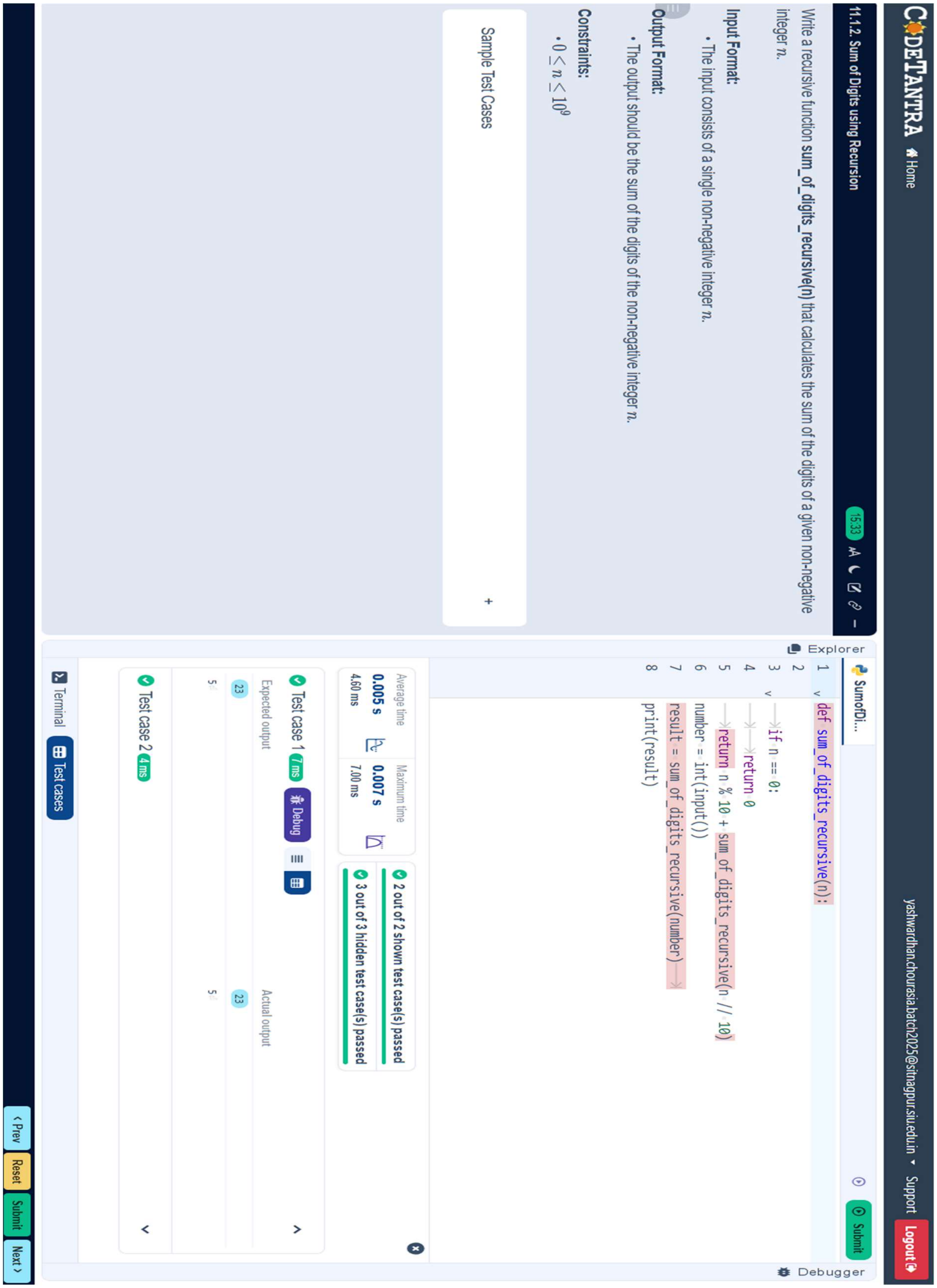
Step 8: End the program

FLOWCHART



Start
Enter n
Call sum_of_digits(n)
n == 0 ?
Return 0
Return (n % 10) + sum_of_digits(n // 10)
Print result
End

PROGRAM



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11.1.2. Sum of Digits using Recursion
15:33
Write a recursive function sum_of_digits_recursive(n) that calculates the sum of the digits of a given non-negative integer n.
Input Format:
• The input consists of a single non-negative integer n.
Output Format:
• The output should be the sum of the digits of the non-negative integer n.
Constraints:
• 0 ≤ n ≤ 10^9
Sample Test Cases
Explorer
SumofDi...
Submit
def sum_of_digits_recursive(n):
if n == 0:
return 0
return n % 10 + sum_of_digits_recursive(n // 10)
number = int(input())
result = sum_of_digits_recursive(number)
print(result)
Average time
0.005 s
4.60 ms
Maximum time
0.007 s
7.00 ms
2 out of 2 shown test case(s) passed
3 out of 3 hidden test case(s) passed
Test case 1
7 ms
Debug
Expected output
23
5
Actual output
23
5
Test case 2
4 ms
Terminal
Test cases
Debugger
Prev
Reset
Submit
Next